

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a height of **8 meters (m)** and a base with a radius of **12 m**. What is the volume, **in m³**, of the cylinder?

Answer

- A. 8π
- B. 20π
- C. 768π
- D. $1,152\pi$

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a right circular cylinder is given by $V = \pi r^2 h$, where r is the radius of the circular base and h is the height of the cylinder. It's given that the cylinder has a height of **8 meters** and a base with a radius of **12 meters**. Substituting **12** for r and **8** for h in $V = \pi r^2 h$ yields $V = \pi(12)^2(8)$, or $V = 1,152\pi$. Therefore, the volume, in **m³**, of the cylinder is **1,152π**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the volume, in **m³**, of a cylinder with a radius of **8 meters** and a height of **12 meters**.